Nicholas Johnson

Computational Neuroscience B.S. | Neuroinformatics | George Mason University

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RESEARCH

Neuroinformatics Research Assistant | NeuroMorpho.Org **June 2026 - Present**
Krasnow Institute for Advanced Study, George Mason University *Fairfax, VA*
Under the supervision of Dr. Carolina Tolama

- Conduct literature mining to identify neuronal and glial reconstructions relevant to NeuroMorpho.Org database curation.
- Evaluate reconstruction evidence using tracing-system keywords including Neurolucida, Imaris/Filament Tracer, Amira, Vaa3D, Simple Neurite Tracer, and CATMAID.
- Curate metadata by verifying article titles, author information, correspondence details, reconstruction counts, figure evidence, supplementary material, and data availability.
- Review figures, captions, supplemental files, and repositories such as GitHub, Dryad, Figshare, Zenodo, and Dataverse for downloadable reconstruction data.
- Document evidence-based curation comments linking reconstruction methods, morphology parameters, data sources, DOI/PMID references, and confidence levels.

Undergraduate Research Assistant | Anatomy and Physiology Education Research **January 2026 - May 2026**
George Mason University *Fairfax, VA*
Under the supervision of Dr. Herin

- Collaborated on manuscript preparation for a comparative anatomy and physiology education study.
- Contributed to data interpretation, scientific writing, editing, and narrative development for peer-reviewed publication.

COMPUTATIONAL PROJECTS

Hodgkin-Huxley Computational Neuroscience Project **Independent Research Project | May 2026 - Present**
Python-based biophysical neuron simulation with NEURON validation

- Designed and implemented a conductance-based neuron simulator to model action potential generation and membrane dynamics.
- Tested how sodium and potassium conductance parameters alter spike morphology and firing behavior through computational experiments.
- Validated simulation outputs against NEURON to evaluate numerical accuracy and biological realism.
- Authored a research-style manuscript documenting methods, results, error analysis, biological interpretation, and future extensions.

EDUCATION

George Mason University **Expected Graduation: Summer 2027**
B.S. Neuroscience, Concentration: Computational Neuroscience *Fairfax, VA*

- Curriculum focused on molecular/cellular neuroscience, computational neuroscience, neural networks, and neural informatics.
- Treasurer, Students in Neuroscience - administering Fall 2026 budget and expense tracking for the student organization.
- Selected as one of 18 from 60 applicants for the GEO Neurotechnology in Germany Program, a 6-credit program visiting 8 research labs across Germany.

Northern Virginia Community College **2023 - 2025**
A.S. Engineering *Annandale Campus, VA*

- Completed general education and core STEM curriculum.

EXPERIENCE

MRI Office Assistant **July 2025 - Present**
Institute for Biohealth Innovation, George Mason University *Fairfax, VA*

- Complete administrative tasks for research administration staff and support MRI office operations.
- Received entry-level MRI training from a certified MRI Technologist.

Office Assistant, IET Department **February 2025 - May 2025**
Northern Virginia Community College *Annandale Campus, VA*

- Created simulation labs for computational computer networks and supported IT lab hardware/network setup.
- Debugged and transferred hardware in student-rental laptops; configured networks and switches for IT labs.

TECHNICAL SKILLS

Computational neuroscience: Hodgkin-Huxley modeling, membrane dynamics, conductance-based simulation, NEURON validation
Neuroinformatics: literature mining, morphology metadata curation, reconstruction evidence review, DOI/PMID linking
Tools and data sources: Python, NEURON, GitHub, Dryad, Figshare, Zenodo, Dataverse, scholarly database review
Scientific communication: manuscript preparation, figure/caption review, data interpretation, evidence-based scientific writing